Trigonometrija

Adicijski teoremi

- $\sin(x \pm y) = \sin x \cos y \pm \cos x \sin y$
- $\cos(x \pm y) = \cos x \cos y \mp \sin x \sin y$
- $\operatorname{tg}(x \pm y) = \dfrac{\operatorname{tg} x \pm \operatorname{tg} y}{1 \mp \operatorname{tg} x \operatorname{tg} y}$
- $\operatorname{ctg}(x \pm y) = \dfrac{\operatorname{ctg} x \operatorname{ctg} y \mp 1}{\operatorname{ctg} y \pm \operatorname{ctg} x}$

Funkcije višestrukih argumenata

- $\sin 2x = 2 \sin x \cos x$
- $\cos 2x = \cos^2 x - \sin^2 x$
- $\operatorname{tg} 2x = \dfrac{2 \operatorname{tg} x}{1 - \operatorname{tg}^2 x}$
- $\operatorname{ctg} 2x = \dfrac{\operatorname{ctg}^2 x - 1}{2 \operatorname{ctg} x}$

Formule pretvorbe

- $\sin x + \sin y = 2 \sin \dfrac{x+y}{2} \cos \dfrac{x-y}{2}$
- $\sin x - \sin y = 2 \cos \dfrac{x+y}{2} \sin \dfrac{x-y}{2}$
- $\cos x + \cos y = 2 \cos \dfrac{x+y}{2} \cos \dfrac{x-y}{2}$
- $\cos x - \cos y = -2 \sin \dfrac{x+y}{2} \sin \dfrac{x-y}{2}$
- $\sin x \cos y = \dfrac{1}{2}(\sin(x+y) + \sin(x-y))$
- $\cos x \cos y = \dfrac{1}{2}(\cos(x+y) + \cos(x-y))$
- $\sin x \sin y = \dfrac{1}{2}(\cos(x-y) - \cos(x+y))$

Funkcije polovičnih argumenata

- $\sin^2 \dfrac{x}{2} = \dfrac{1 - \cos x}{2}$
- $\cos^2 \dfrac{x}{2} = \dfrac{1 + \cos x}{2}$

Tablica derivacija

- $(x^c)' = cx^{c-1} \quad (c \in \mathbb{R},\, x > 0)$
- $(|x|)' = \dfrac{x}{|x|} \quad (x \neq 0)$
- $(e^x)' = e^x$
- $(a^x)' = a^x \ln a \quad (a > 0)$
- $(\ln x)' = \dfrac{1}{x} \quad (x > 0)$
- $(\log_a x)' = \dfrac{1}{x \ln a} \quad (a > 0,\, a \neq 1,\, x > 0)$
- $(\sin x)' = \cos x$
- $(\arcsin x)' = \dfrac{1}{\sqrt{1 - x^2}} \quad (|x| < 1)$
- $(\cos x)' = -\sin x$
- $(\arccos x)' = -\dfrac{1}{\sqrt{1 - x^2}} \quad (|x| < 1)$
- $(\operatorname{tg} x)' = \dfrac{1}{\cos^2 x}$
- $(\operatorname{arctg} x)' = \dfrac{1}{1 + x^2}$
- $(\operatorname{ctg} x)' = -\dfrac{1}{\sin^2 x}$
- $(\operatorname{arcctg} x)' = -\dfrac{1}{1 + x^2}$

- $(\operatorname{sh} x)' = \operatorname{ch} x$
- $(\operatorname{ch} x)' = \operatorname{sh} x$
- $(\operatorname{th} x)' = \dfrac{1}{\operatorname{ch}^2 x}$
- $(\operatorname{cth} x)' = -\dfrac{1}{\operatorname{sh}^2 x}$
- $(\operatorname{arsh} x)' = \dfrac{1}{\sqrt{1+x^2}}$
- $(\operatorname{arch} x)' = \dfrac{1}{\sqrt{x^2-1}} \quad (x > 1)$
- $(\operatorname{arth} x)' = \dfrac{1}{1-x^2} \quad (|x| < 1)$
- $(\operatorname{arcth} x)' = \dfrac{1}{1-x^2} \quad (|x| > 1)$

Tablica integrala

- $\displaystyle\int x^a\,dx = \frac{x^{a+1}}{a+1} + C \quad (a \neq -1)$
- $\displaystyle\int e^x\,dx = e^x + C$
- $\displaystyle\int \cos x\,dx = \sin x + C$
- $\displaystyle\int \frac{dx}{\cos^2 x} = \operatorname{tg} x + C$
- $\displaystyle\int \operatorname{ch} x\,dx = \operatorname{sh} x + C$
- $\displaystyle\int \frac{dx}{\operatorname{ch}^2 x} = \operatorname{th} x + C$
- $\displaystyle\int \frac{dx}{\sqrt{a^2-x^2}} = \arcsin\frac{x}{a} + C \quad (a > 0)$
- $\displaystyle\int \frac{dx}{x^2+a^2} = \frac{1}{a}\operatorname{arctg}\frac{x}{a} + C \quad (a > 0)$
- $\displaystyle\int \frac{dx}{x} = \ln|x| + C$
- $\displaystyle\int a^x\,dx = \frac{a^x}{\ln a} + C \quad (a > 0,\ a \neq 1)$
- $\displaystyle\int \sin x\,dx = -\cos x + C$
- $\displaystyle\int \frac{dx}{\sin^2 x} = -\operatorname{ctg} x + C$
- $\displaystyle\int \operatorname{sh} x\,dx = \operatorname{ch} x + C$
- $\displaystyle\int \frac{dx}{\operatorname{sh}^2 x} = -\operatorname{cth} x + C$
- $\displaystyle\int \frac{dx}{\sqrt{x^2+a}} = \ln|x + \sqrt{x^2+a}| + C \quad (a \neq 0)$
- $\displaystyle\int \frac{dx}{x^2-a^2} = \frac{1}{2a}\ln\left|\frac{x-a}{x+a}\right| + C \quad (a > 0)$

Tablica limesa

- $\displaystyle\lim_{x\to 0}\frac{\sin x}{x} = 1$
- $\displaystyle\lim_{x\to 0}\frac{e^x - 1}{x} = 1$
- $\displaystyle\lim_{x\to 0}\frac{\ln(1+x)}{x} = 1$
- $\displaystyle\lim_{x\to\pm\infty}\left(1+\frac{1}{x}\right)^x = e$

Napomena: Na službeni podsjetnik strogo je zabranjeno bilo što dopisivati. Također, zabranjeno je i tijekom samog pisanja ispita pisati po ovom službenom podsjetniku. Čuvari će tijekom pisanja ispita obići dvoranu i ako primijete bilo kakav dodatak na službenom podsjetniku, dotični student će biti udaljen s ispita.